# A Complete Guide to Parallel Testing

**Setting Up Your Selenium Grid 4 Environment**

# Contents

# 1 Introduction

This guide provides a complete, step-by-step process for using the pre-configured Selenium Grid. This project is all-inclusive: the server `.jar`, all drivers, and test code are provided.

The goal is to have a robust, fast, and repeatable testing environment that anyone on the team can use immediately.

# 2 Part 1: Selenium Grid Setup

The Grid consists of one "Hub" (the manager) and multiple "Nodes" (the workers). The project includes batch files to start them easily.

## 2.1 Project Structure and Paths

The project is pre-configured to work with this exact folder structure. The `.bat` files rely on these paths.

1. **Server and Test Code:** All `.bat` files, the `selenium-server-4.38.0.jar` file, and the Python test code are located in:

   ```
   D:/Selenium_Grid/
   ```

2. **WebDrivers (Chrome & Gecko):** The included drivers (`chromedriver.exe` and `geckodriver.exe`) are located in:

   ```
   C:/WebDriver/bin/
   ```

   If your system paths are different, you will need to edit the `.bat` files to match your locations.

## 2.2 Understanding the Batch (.bat) Files

The `D:/Selenium_Grid/` folder contains three `.bat` files to run the Grid. Here is what each file does.

### 2.2.1 1. `start_hub.bat`

This file starts the main Hub, which manages all the nodes.

```
java -jar D:\Selenium_Grid\selenium-server-4.38.0.jar hub
```

### 2.2.2 2. `start_node_chrome.bat`

This file starts a Chrome Node and connects it to the Hub. It uses a specific port (`5556`) and connects to the Hub's event bus for more stable communication.

```
java -Dwebdriver.chrome.driver=C:\WebDriver\bin\chromedriver.exe ^
-jar D:\Selenium_Grid\selenium-server-4.38.0.jar node ^
--port 5556 ^
--publish-events tcp://192.168.18.38:4442 ^
--subscribe-events tcp://192.168.18.38:4443
```

### 2.2.3 3. `start_node_firefox.bat`

This file does the same, but for Firefox, using port `5557`.

```
java -Dwebdriver.gecko.driver=C:\WebDriver\bin\geckodriver.exe ^
-jar D:\Selenium_Grid\selenium-server-4.38.0.jar node ^
--port 5557 ^
--publish-events tcp://192.168.18.38:4442 ^
--subscribe-events tcp://192.168.18.38:4443
```

## 2.3 Running the Grid

This is the easy part!

1. **Run** `start_hub.bat`. A command prompt will open. Wait until you see a message like "Started Selenium hub...".
2. **Open the Grid UI:** In your browser, go to `http://localhost:4444`. You will see the Grid dashboard.
3. **Run** `start_node_chrome.bat`.
4. **Run** `start_node_firefox.bat`.
5. **Check the UI:** Refresh the Grid dashboard. You will now see your Chrome and Firefox nodes attached, ready to receive tests!

# 3 Part 2: Python Test Setup

Now we set up the Python environment to *run* the tests.

## 3.1 Install Python Packages

The project includes a `requirements.txt` file to install all necessary Python packages.

1. Open a **new** command prompt or terminal.
2. Navigate to your project folder:

```
cd D:/Selenium_Grid/
```

3. Install all the required packages at once by running:

```
pip install -r requirements.txt
```

# 4 Part 3: Run the Tests and See the Magic!

With the Grid running and your Python packages installed, you are ready.

## 4.1 The Magic Command

From your `D:/Selenium_Grid/` directory, run this command:

```
python -m pytest -n auto
```

## 4.2 What's Happening

- **`python -m pytest`** finds all `test_*.py` files in the `test_cases` folder.
- The `conftest.py` file tells `pytest` to run *every* test twice: once for Chrome, once for Firefox.
- **`-n auto`** (from the `pytest-xdist` package) tells `pytest` to run all these tests in parallel, using all available CPU cores.
- You can specify a number, e.g., `-n 4` to run 4 tests at once.
- **Watch the Grid UI!** You will see the tests get assigned to your nodes in real-time, and multiple browsers will pop up and run simultaneously.

# 5 Final Notes

**Understanding Paths:** The batch files are hard-coded to use the paths specified in this guide (e.g., `D:/Selenium_Grid/` and `C:/WebDriver/bin/`). If you move the project or drivers, you MUST update the paths in the `.bat` files accordingly.

That's it! You now have a complete, high-performance parallel testing setup.